

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Integrates using integration by parts	AO3.1a	M1	$y = \int (x-1)e^x dx$
Applies integration by parts formula correctly to either of $(x-1)e^x$ or xe^x	AO1.1a	M1	$u = x-1 \quad \frac{dv}{dx} = 1$ $\frac{dv}{dx} = e^x \quad v = e^x$
Obtains fully correct integral, condone missing constant.	AO1.1b	A1	$y = (x-1)e^x - \int e^x dx$ $y = (x-1)e^x - e^x + c$
Explains clearly why the minimum y value is e with reference to the range of the function OE	AO2.4	E1	Range $\geq e \Rightarrow$ at min $y = e$ Min point when $\frac{dy}{dx} = 0 \therefore x = 1$
Uses $\frac{dy}{dx} = 0$ to find x coordinate of minimum	AO1.1a	M1	So curve passes through $(1, e)$ $e = (1-1)e^1 - e^1 + c$
Deduces that the curve passes through the point $(1, e)$	AO2.2a	A1	$c = 2e$
Uses their minimum point to find their c	AO1.1a	M1	$\therefore f(x) = (x-2)e^x + 2e$
States the correct equation in any correct form Condone y instead of $f(x)$ CAO	AO1.1b	A1	
			Total 8 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving $\frac{dx}{dt}$, t , x and a constant of proportionality.	AO3.3	B1	$\frac{dx}{dt} = \frac{k(8-t)}{x}$

	Substitutes $t = 2$, $x = 336$, $\frac{dx}{dt} = 72$ to find k	AO1.1a	M1	$72 = \frac{k(8-2)}{336}$ $k = 4032$
	Obtains correct value of k shows given result AG	AO2.1	R1	$\frac{dx}{dt} = \frac{4032(8-t)}{x}$ $x \frac{dx}{dt} = 4032(8-t)$
(b)	Integrates one side correctly	AO1.1a	M1	$\int x dx = \int 4032(8-t) dt$
	Integrates both sides correctly condone missing c	AO1.1b	A1	$\frac{1}{2}x^2 = 4032\left(8t - \frac{t^2}{2}\right) + c$
	Uses conditions to show $c = 0$ and correctly obtains given result. AG	AO2.1	R1	$\frac{1}{2} \times 336^2 = 4032\left(8 \times 2 - \frac{2^2}{2}\right) + c$ $c = 0$ $x^2 = 64512t - 4032t^2$ $\therefore x^2 = 4032t(16-t)$
(c)	Translates rate of growth into $\frac{dx}{dt} = 24$ and uses in model for rate of sales	AO3.3	M1	$24x = 4032(8-t)$ $x = 168(8-t)$ $(168(8-t))^2 = 4032t(16-t)$
(i)	Eliminates x to form quadratic equation or inequality in t	AO3.1a	M1	$t^2 - 16t + 56 = 0$ $t = 5.171.. \text{ or } 10.828...$
	Obtains correct equation or inequality in t Need not be simplified	AO1.1b	A1	5 hours 10 minutes
	Obtains $t = 5.17$	AO1.1b	A1	Earliest time 14:40
	Converts their t into hours and minutes	AO3.4	A1	
	Interprets the closing time as 14:40 09:30 + their converted time	AO3.2a	R1F	
(ii)	Explains in context that when the stall opens there will be zero sales	AO3.5a	E1	When the stall opens there are zero sales
	Explains that when $x = 0$ the model is undefined	AO3.5a	E1	$\frac{dx}{dt}$ When $x = 0$ $\frac{dx}{dt}$ is undefined as the denominator is zero

Total 14 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Separates variables, at least one side correct.	AO3.1a	M1	$3\sqrt{x} \frac{dx}{dt} = 8 \sin 2t$
	Obtains correct separation PI	AO1.1b	A1	$\int 3\sqrt{x} dx = \int 8 \sin 2t dt$
	integrates 'their' expressions, at least one of 'their' sides correct	AO1.1a	M1	$\int 3x^{\frac{1}{2}} dx = \int 8 \sin 2t dt$
	Obtains correct integral (condone missing + c) CAO	AO1.1b	A1	$2x^{\frac{3}{2}} = -4 \cos 2t (+c)$
	Substitutes initial conditions, to find + c.	AO3.1b	M1	$2 \times (0)^{\frac{3}{2}} = -4 \cos(2 \times 0) + c$ $c = 4$
	Obtains a correct solution ACF	AO1.1b	A1	$x^{\frac{3}{2}} = 2 - 2 \cos 2t$
	Obtains correct solution of the form $x = f(t)$	AO2.5	A1	$x = (2 - 2 \cos 2t)^{\frac{2}{3}}$
(b)	Obtains correct max height, in cm Award FT from correct substitution into incorrect equation $x = f(t)$ but only if all three M1 marks have been awarded, must have correct units.	AO3.4	A1F	max height $= 4^{\frac{2}{3}} = 252 \text{ cm}$
Total 8 marks				

Q4.

(a) $\int x(x^2 + 3)^{\frac{1}{2}} dx = p(x^2 + 3)^{\frac{3}{2}}$

By inspection or substitution

M1

$$= \frac{1}{3}(x^2+3)^{\frac{3}{2}} (+C)$$

A1

2

(b) $\int e^{2y} dy = \int x\sqrt{x^2+3} dx$

Correct separation and notation

Condone missing integral signs

B1

$$= \frac{1}{2}e^{2y}$$

B1

$$= \frac{1}{3}(x^2+3)^{\frac{3}{2}} + C$$

Equate to result from (a) with constant.

M1

$$\frac{1}{2} = \frac{1}{3} \times 4^{\frac{3}{2}} + C$$

Use (1,0) to find constant.

m1

$$C = -\frac{13}{6}$$

CAO

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

A1

$$2y = \ln \left(\frac{2}{3}(x^2+3)^{\frac{3}{2}} - \frac{13}{3} \right)$$

Solve for y, taking logs correctly.

m1

$$y = \frac{1}{2} \ln \left(\frac{2}{3}(x^2+3)^{\frac{3}{2}} - \frac{13}{3} \right)$$

CSO

A1

7

[9]

Q5.

(a) $\int t \cos\left(\frac{\pi}{4}t\right) dt$

Clear attempt to use parts

$$u = t \quad \frac{dv}{dt} = \cos\left(\frac{\pi}{4}t\right)$$

$$\frac{du}{dt} = 1 \quad v = k \sin\left(\frac{\pi}{4}t\right)$$

M1

$$= t \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}t\right) - \frac{4}{\pi} \int \sin\left(\frac{\pi}{4}t\right) (dt)$$

Must be in terms of π

A1

$$= pt \sin\left(\frac{\pi}{4}t\right) + q \cos\left(\frac{\pi}{4}t\right)$$

Correct form, any non-zero values for p, q

m1

$$= t \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}t\right) + \frac{4}{\pi} \times \frac{4}{\pi} \cos\left(\frac{\pi}{4}t\right)$$

Any correct unsimplified form.

Constant not required

A1

4

(b) $\int 32x \, dx = \int t \cos\left(\frac{\pi}{4}t\right) dt$

Correct separation and notation.

B1

$$16x^2 =$$

$$\frac{x^2}{2} \text{ if 32 not brought over; allow } 32 \times \frac{x^2}{2}$$

B1

$$t \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}t\right) + \frac{16}{\pi^2} \cos\left(\frac{\pi}{4}t\right) + C$$

Equate to result from part (a) with constant and use (0,4) to find a value for the constant

M1

$$C = 256 - \frac{16}{\pi^2}$$

Accept $C = 254$ or better (254.37886...)

A1

$$t = 45$$

$$16x^2 = -40.514... - 1.146... + 254.378... = 212.718...$$

$$x^2 = 13.294...$$

$$x = 3.646... = 3.65 \text{ m or (height =) } 365 \text{ cm}$$

$$\text{Substitute } t = 45 \text{ into } kx^2 = p t \sin\left(\frac{\pi}{4}t\right) + q \cos\left(\frac{\pi}{4}t\right) + C$$

$p \neq 0$, $q \neq 0$ and calculate x .

CSO

m1A1

6

[10]

Q6.

$$\int \frac{dy}{y^2} = \int x \sin 3x \, dx$$

Correct separation and notation;
condone missing integral signs

B1

$$\int \frac{dy}{y^2} = -\frac{1}{y}$$

©2025 Exam Papers Practice. All rights reserved.

B1

$$\int x \sin 3x \, dx = x \left(-\frac{1}{3} \cos 3x \right)$$

$$u = x \quad \frac{dv}{dx} = \sin 3x$$

Use parts $\frac{du}{dx} = 1$ $v = k \cos 3x$ with correct substitution into formula

$$- \int -\frac{1}{3} \cos 3x$$

A1

$$= -\frac{1}{3}x \cos 3x \, dx + \frac{1}{9} \sin 3x$$

CAO

A1

$$-\frac{1}{y} = -\frac{1}{3}x \cos 3x + \frac{1}{9} \sin 3x + C$$

$$-1 = -\frac{1}{3} \times \frac{\pi}{6} \cos\left(\frac{\pi}{2}\right) + \frac{1}{9} \sin\left(\frac{\pi}{2}\right) + C$$

$$\text{Use } x = \frac{\pi}{6} \, y = 1 \text{ to find } C$$

M1

$$C = -\frac{10}{9}$$

CAO

A1

$$-\frac{1}{y} = -\frac{1}{9} (3x \cos 3x \, dx - \sin 3x + 10)$$

$$\text{And invert to } -y = -\frac{9}{(\dots)}$$

m1

$$y = \frac{9}{3x \cos 3x - \sin 3x + 10}$$

CSO, condone first B1 not given

A1

Second M1 finding C; substitute $x = \frac{\pi}{6} \, y = 1$ into $f(y) = px \cos 3x + q \sin 3x + C$ and evaluate using radians.

Must calculate a value of C.

$$m1 \text{ for reaching form } \pm \frac{k}{y} = \frac{1}{9} (Px \cos 3x + Q \sin 3x + R) \text{ where } P \text{ and } Q \text{ are } \pm 3 \text{ or } \pm \frac{1}{3}$$

$$\text{or } \pm 1 \text{ and inverting to } \pm \frac{y}{k} = \frac{9}{(Px \cos 3x + Q \sin 3x + R)}$$

[9]

Q7.

(a) $\frac{dh}{dt}$

B1

derivative = * $\times (2 - h)$

Use of $2 - h$ or $h - 2$;

* is a constant or expression in h and / or t .

M1

$$\frac{dh}{dt} = k(2 - h)$$

All correct; must be $(2 - h)$

A1

3

(b) (i) $\int x\sqrt{2x-1} \, dx = \int \frac{1}{15} \, dt$

Correct separation and notation;
condone missing integral signs.

B1

$$= \frac{1}{15} t$$

B1

Substitute $u = 2x - 1$

$$\int x\sqrt{2x-1} \, dx = \int \frac{1}{2}(u+1)\sqrt{u} \frac{1}{2} du$$

Suitable substitution and attempt to write integral in terms
of u including dx replaced by $\frac{1}{2} du$ or $2du$.

M1

$$= \left(\frac{1}{4}\right) \int u^{\frac{3}{2}} + u^{\frac{1}{2}} \, du$$

$\frac{1}{4}$ need not be seen

A1

$$= \frac{1}{4} \left(\frac{2}{5} u^{\frac{5}{2}} + \frac{2}{3} u^{\frac{3}{2}} \right) (+ C)$$

Integration correct including $\frac{1}{4}$

A1

$$x = 1; t = 0$$

$$u = 1, t = 0 \quad \frac{1}{4} \left(\frac{2}{5} + \frac{2}{3} \right) + C = 0$$

Use $x = 1, t = 0$ to find a value for constant C from equation in x and t .

M1

$$C = -\frac{4}{15}$$

$$C = -0.2666\dots$$

$$C = -0.267 \text{ or better}$$

A1

$$t = \frac{1}{2} \left(3(2x-1)^{\frac{5}{2}} + 5(2x-1)^{\frac{3}{2}} \right) - 4$$

$$\text{ISW } t = (2x-1)^{\frac{3}{2}} (3x+1) - 4$$

A1

Alternative (Parts)

As before

(B1B1)

$$u = x, \quad \frac{dv}{dx} = (2x-1)^{\frac{1}{2}}$$

Attempt to use parts

©2025 Exam Papers Practice. All rights reserved. (M1)

$$du = 1 \quad v = k(2x-1)^{\frac{3}{2}}$$

$$\int x\sqrt{2x-1} \, dx = x \frac{1}{3} (2x-1)^{\frac{3}{2}} - \int \frac{1}{3} (2x-1)^{\frac{3}{2}} \, dx$$

Condone missing dx

(A1)

$$= x \frac{1}{3} (2x-1)^{\frac{3}{2}} - \frac{1}{15} (2x-1)^{\frac{5}{2}} (+C)$$

(A1)

$$x = 1, t = 0 \quad \frac{1}{3} - \frac{1}{15} + C = 0$$

Use $x = 1, t = 0$ to find a value for constant C from equation in x and t

(M1)

$$C = -\frac{4}{15}$$

$$C = -0.2666\dots$$

$$C = -0.267 \text{ or better}$$

(A1)

$$t = 5x(2x-1)^{\frac{3}{2}} - (2x-1)^{\frac{5}{2}} - 4$$

$$\text{ISW } t = (2x-1)^{\frac{3}{2}}(3x+1) - 4$$

(A1)

8

(ii) $x = 2 \quad t = 32.4 \text{ (minutes)}$
 $32.4 \text{ or better } (32.373\dots)$

B1

1

[12]

Q8.

(a) (i) $\int \frac{dx}{\sqrt{x}} = \int \sin\left(\frac{t}{2}\right) dt$

Correct separation; condone missing integral signs.

B1

$$2\sqrt{x} = -2\cos\left(\frac{t}{2}\right) (+ k)$$

$$p\sqrt{x} = q\cos\left(\frac{t}{2}\right)$$

Condone missing $+ k$

M1

$$x = \left(-\cos\left(\frac{t}{2}\right) + C \right)^2$$

Must have previous line correct

A1

3

(ii) $(1, 0) \quad 2 = -2 + k \text{ or } 1 = (-1 + C)^2$

Use (1, 0) to find a constant

M1

$$k = 4 \text{ or } C = 2$$

ft on $C = p - q$ from (a)(i)

A1ft

$$x = \left(2 - \cos\left(\frac{t}{2}\right) \right)^2$$

cso applies to (a)(ii)

A1

3

(b) (i) Greatest height when $\cos(bt) = -1$

M1

Greatest height = 9 (m)

ft is (their $a + 1$)²

A1ft

2

(ii) $\cos\left(\frac{t}{2}\right) = 2 - \sqrt{5}$

$$\cos bt = a - \sqrt{5}$$

M1

$$t = 2 \cos^{-1}(2 - \sqrt{5}) = 3.6 \text{ (seconds 1dp)}$$

condone 3.6 or better (3.618.....)

A1

2

[10]

Q9.

(a) $\frac{dA}{dt}$

B1

$$= -k$$

B1

2

(b) (i) $A = -kt (+ C)$
Integrate

M1

$$C = 4\pi \times 60^2$$

$$C \text{ correct from } A = \pm k + C$$

A1

$$4\pi \times 30^2 = -9k + 4\pi \times 60^2$$

*Use $r = 30$ $t = 9$ and attempt to find k ,
 as far as $k = \dots$*

$$k = 1200\pi$$

m1

$$A = -1200\pi t + 14400\pi$$

$$= 1200\pi(12 - t)$$

AG CSO

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

A1

4

(ii) $t = 12$ (days)

B1

1

[7]

Q10.

(a) $1 = A(1 - x)^2 + B(1 - x)(3 - 2x) + C(3 - 2x)$
Attempt to clear fractions

M1

$$\left. \begin{array}{lll} x = 1 & x = \frac{3}{2} & x = 0 \\ C = 1 & 1 = A\left(-\frac{1}{2}\right)^2 & 1 = A + 3B + 3C \end{array} \right\}$$

Use any two (or three) values of x to set up two (or three) equations

m1

$$A = 4 \quad B = -2 \quad C = 1$$

Two values correct

A1

All values correct

A1

Alternative

$$1 = A(1 - x)^2 + B(1 - x)(3 - 2x) + C(3 - 2x)$$

(M1)

$$1 = A + 3B + 3C$$

$$0 = -2A - 5B - 2C$$

Set up three simultaneous equations

(m1)

$$0 = A + 2B$$

Two values correct

(A1)

$$A = 4 \quad B = -2 \quad C = 1$$

All values correct

(A1)

4

(b)
$$\int \frac{1}{2\sqrt{y}} dy = \int \frac{4}{3-2x} - \frac{2}{1-x} + \frac{1}{(1-x)^2} dx$$

Separate using partial fractions; correct notation;
condone missing integral signs but dy and dx
must be in correct place.

ft on their A , B , C and on each integral.

B1ft

$$\int \frac{1}{2\sqrt{y}} dy = \sqrt{y} =$$

OE $\int \frac{k}{\sqrt{y}} dy = 2k\sqrt{y}$ is B1

B1

$$- 2 \ln (3 - 2x)$$

Condone missing brackets on one ln integral.

B1ft

$$+ 2 \ln (1 - x)$$

B1ft

$$+ \frac{1}{1-x} (+C)$$

Condone omission of +C

B1ft

$$x = 0 \quad y = 0 \Rightarrow 0 = -2 \ln 3 + 0 + 1 + C$$

Use (0,0) to find C. Must get to C =

M1

$$C = 2 \ln 3 - 1$$

Correct C found from correct equation.

C must be exact, in any form but not decimal.

A1

$$\sqrt{y} = 2 \ln \left(\frac{3-3x}{3-2x} \right) + \frac{1}{1-x} - 1$$

Correct use of rules of logs to progress towards requested form of answer. C must be of the form $r \ln s + t$

m1

$$y^{\frac{1}{2}} = 2 \ln \left(\frac{3-3x}{3-2x} \right) + \frac{x}{1-x}$$

OE

CSO condone B0 for separation

A1

9

[13]

Q11.

$$\int y \, dy = \int \cos\left(\frac{x}{3}\right) \, dx$$

Separate; condone missing integral signs.

B1

$$\frac{1}{2} y^2 = 3 \sin\left(\frac{x}{3}\right) + (C)$$

$$\frac{\sin\left(\frac{x}{3}\right)}{\frac{1}{3}}$$
 Accept

B1B1

$$\left(\frac{\pi}{2}, 1\right) \quad \frac{1}{2} = 3 \sin \frac{\pi}{6} + C$$

Use $\left(\frac{\pi}{2}, 1\right)$ to find C

must be in form $py^2 = q \sin\left(\frac{x}{3}\right) + C$

EXAM PAPERS PRACTICE

M1

$$C = -1$$

©2025 Exam Papers Practice. All rights reserved.

A1F

$$y^2 = 6 \sin\left(\frac{x}{3}\right) - 2$$

CSO

A1

[6]

Q12.

(a)
$$\int \frac{dx}{\sqrt{x+1}} = \int -\frac{1}{5} \, dx$$

$\frac{dt}{dx} = -5(x+1)^{\frac{1}{2}}$
 Correct separation; or
 Condone missing integral signs

B1

$$2\sqrt{x+1} = -\frac{1}{5}t \quad (+C)$$

Correct integrals; condone
 $\frac{\sqrt{x+1}}{\frac{1}{2}}$

B1B1

$$x = 80 \quad t = 0 \quad C = 2\sqrt{81}$$

Use (0, 80) to find a constant C

M1

$$= 18$$

F on integrals if in form $\sqrt{x+1} = qt + c$

A1F

$$x = \left(9 - \frac{1}{10}t\right)^2 - 1$$

OE; CSO; x = correct expression in t

A1

EXAM PAPERS PRACTICE

6

(b) $t = 60 \quad x = f(60)$

Evaluate $f(60)$, ie $x = \dots$ (C not required)

M1

$$= 8$$

CSO

A1

2

(c) (i) $\frac{dA}{dt} = kA(9 - A)$

$\frac{dA}{dt}$ = product involving A; k required
 Condone terms in t

M1A1

2

$$(ii) \quad 4.5 = \frac{9}{1 + 4e^{-0.09t}}$$

Condone one slip in denominator

M1

$$e^{-0.09t} = \frac{1}{4}$$

A1

$$-0.09t = \ln\left(\frac{1}{4}\right)$$

Take ln correctly

m1

$$t = \frac{\ln\left(\frac{1}{4}\right)}{-0.09}$$

$$= 15.4 \text{ (hours)}$$

CAO; condone more than 3sf if correct
15.40327068 Allow 15h 24m

A1

4

[14]

Q13.

$$(a) \quad (i) \quad \int \frac{dx}{e^{\frac{1}{2}x}} = \int -kt \, dt$$

Separate; condone missing integral signs

B1

$$-2e^{-\frac{1}{2}x} = -k \frac{t^2}{2} \quad (+C)$$

B1B1

3

$$(ii) \quad -2e^{-\frac{1}{2}x} = -k \frac{t^2}{2} - 2e^{-3}$$

Use (6, 0) to find constant

M1

$$\ln \left(e^{\frac{1}{2}x} \right) = \ln \left(k \frac{t^2}{4} + e^{-3} \right)$$

Take logarithms correctly; condone one side negative. Must have a constant.

M1

$$-\frac{1}{2}x = \ln \left(k \frac{t^2}{4} + e^{-3} \right)$$

$$x = -2 \ln \left(\frac{kt^2}{4} + e^{-3} \right)$$

Answer given; CSO

A1

3

(b) (i) $t = 10 \quad x = -2 \ln \left(\frac{0.004 \times 10^2}{4} + e^{-3} \right)$

$= 3.8 \Rightarrow 3800$
CAO

M1

A1

2

(ii) $x = 0 \quad \frac{0.004 \times t^2}{4} + e^{-3} = 1$

$t = 30.8$

CAO

Treat 0.04 or 0.0004 as misread (– 1)

M1

A1

2

[10]

Q14.

(a) $\int x \, dx = \int 150 \cos 2t \, dt$

Correct separation; condone missing $\int$ signs; must see dx, dt

B1

$$\frac{1}{2}x^2 = 75 \sin 2t \quad (+C)$$

Correct integrals

Accept $\frac{1}{2} \times 150$

B1B1

$$\left(20, \frac{\pi}{4}\right) \quad \frac{1}{2} \times 20^2 = 75 \sin\left(2 \times \frac{\pi}{4}\right) + C$$

C present. Use $\left(20, \frac{\pi}{4}\right)$ to find C

M1

$$C = 125$$

F on $x^2 = k \sin 2t$

A1F

$$x^2 = 150 \sin 2t + 250$$

Correct integrals and evaluation of C

A1

6

(b) (i) $t = 13 \quad x^2 = 150 \sin 26 + 250 (= 364.38)$

*Evaluate $x^2 = f(13)$; $x^2 = k \sin 2t + c$
with numerical k and t*

M1

$$x = 19.1 \text{ (cm)}$$

AWRT

A1

2

$$(ii) \quad \left. \begin{array}{l} x = 11 \quad \sin 2t = -\frac{129}{150} \quad (= -0.86) \\ \text{or} \quad 2t = -1.035 \dots, 4.176 \dots \end{array} \right\}$$

M1

$$t = 2.1 \text{ (seconds)}$$

AWRT

A1

Q15.

$$\int y \, dy = \int 3 \cos 3x \, dx$$

attempt to separate and integrate
 $py^2 = q \sin 3x$ seen $\Rightarrow$ implies separation

M1

$$\frac{1}{2} y^2 = \sin 3x \quad (+C)$$

integrals – accept $\frac{1}{3} \times 3 \sin 3x$

A1A1

$$\left(\frac{\pi}{2}, 2\right) \frac{1}{2} \times 4 = \sin \frac{3\pi}{2} + C$$

use $\left(\frac{\pi}{2}, 2\right)$ to find constant

M1

$$C = 3$$

$$y^2 = 2 \sin 3x + 6$$

CSO (in any correct form)

A1

©2025 Exam Papers Practice. All rights reserved.

[5]

Q16.

$$(a) \quad \frac{2}{(x^2 - 1)} = \frac{A}{x - 1} + \frac{B}{x + 1}$$

$$2 = A(x + 1) + B(x - 1)$$

M1

$$x = 1$$

$$x = -1$$

use two values of x or equate coefficients and solve $A + B = 0$ and $A - B = 2$

m1

$$A = 1$$

$$B = -1$$

both A and B

A1

3

(b) $\int \frac{2}{x^2 - 1} dx = p \ln(x - 1) + q \ln(x + 1)$

In integrals

M1

$= \ln(x - 1) - \ln(x + 1)$

F on A and B

condone missing brackets

A1F

2

(c) $\int \frac{dy}{y} = \int \frac{2}{3(x^2 - 1)} dx$

separate and attempt to integrate on one side

M1

$\ln y = \frac{1}{3} (\ln(x - 1) - \ln(x + 1)) + C$

left hand side

A1

F from part (b) on right hand side

A1F

(3, 1) $\ln 1 = \frac{1}{3} (\ln 2 - \ln 4) + C$

use (3, 1) to attempt to find a constant

m1

$3 \ln y = \ln(x - 1) - \ln(x + 1) - (\ln 2 - \ln 4)$

$3 \ln y = \ln \left(\frac{x - 1}{x + 1} \right) + \ln 2$

$\ln y^3 = \ln \left(\frac{2(x - 1)}{x + 1} \right)$

$y^3 = \frac{2(x - 1)}{x + 1}$

CSO AG

A1

Q17.

(a) (i) $\int \frac{dy}{y} = \int \sin t \, dt$

Attempt to separate and integrate

M1

$$\ln y = -\cos t + c$$

A1 for $\ln y$; A1 for $-\cos t$; condone missing c

A1,A1

$$y = Ae^{-\cos t}$$

A present; or $y = e^{-\cos t + C}$

A1

4

(ii) $y = 50, t = \pi: 50 = Ae^{-\cos \pi} = Ae$

Substitute $y = 50, t = \pi$ to find constant

Can have $50 = e^{1+C}$ if substituted in above

$$e^C = \frac{50}{e}$$

M1

A1

$$y = 50e^{-1}e^{-\cos t}$$

AG (convincingly obtained)

A1

3

Alternative:

Must have a constant in answer to (a)(i)

$$y = Ae^{-\cos t} \text{ or } y = e^{-\cos t + c} \text{ or } \ln y = -\cos t + c$$

$$50 = Ae^{-\cos \pi} \quad 50 = e^{-\cos \pi + c} \quad \ln 50 = -\cos \pi + c$$

(M1)

$$50 = Ae^{-1} \quad 50 = e^{1+c} \quad \ln y = -\cos t + \ln 50 - 1$$

(A1)

$$y = 50e^{-1-\cos t} \quad y = e^{-\cos t} \quad \frac{50}{e} \ln\left(\frac{y}{50}\right) = -1 - \cos t$$

(A1)

Alternative:

Substitute $y = 50$, $t = \pi$ into $\ln y = -\cos t + c$ M1

$\ln y = -\cos t + \ln 50 - 1$ A1

$$\ln \frac{y}{50} = -1 - \cos t \quad (AG) \quad A1$$

(b) (i) $t = 6: y = 50e^{-1-\cos 6} = 7.0417... \approx 7\text{cm}$

Degrees 6.8 SC1

7 or 7.0 for A1

M1A1

2

(ii) $t = \pi \Rightarrow (\sin t = 0 \Rightarrow) \frac{dy}{dt} = 0$

Condone x for t

B1

$$\frac{d^2 y}{dt^2} = y \cos t + \frac{dy}{dt} \sin t$$

For attempt at product rule including $\frac{dy}{dt}$

term; must have $\frac{d^2 y}{dt^2} =$

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved. M1

$$t = \pi$$

A1

$$\frac{d^2 y}{dt^2} = y \cos \pi + \frac{dy}{dt} \sin \pi$$

$$= -50 \Rightarrow \max$$

Accept $= -y$, with explanation that y is never negative

A1

4

Alternative:

$$y = 50e^{-(1+\cos t)} = \frac{50}{e} e^{-\cos t}$$

$$\frac{dy}{dt} = \frac{50}{e} e^{-\cos t} \times \sin t = 0 \text{ at } t = \pi$$

(B1)

$$\frac{d^2y}{dt^2} = \frac{50}{e} e^{-\cos t} \times \cos t + \frac{50}{e} e^{-\cos t} \times \sin^2 t$$

Attempt at product rule

(M1)

Correct

(A1)

Substitute $t = \pi \rightarrow -50 \Rightarrow \max$

(A1)

[13]

Q18.

(a) $\int \frac{1}{\sqrt{1+2y}} dy = \int \frac{1}{x^2} dx$
attempt to separate and integrate

M1

$\int \frac{1}{\sqrt{1+2y}} dy = k\sqrt{1+2y}$

m1

$$\sqrt{1+2y} = -\frac{1}{x} (+c)$$

OE A1 for $\sqrt{1+2y}$ depends on both Ms

A1

A1 for $-\frac{1}{x}$ depends on first M1 only

A1

$$x = 1, y = 4 \Rightarrow c = 4$$

+ c must be seen on previous line

m1

ft on k and $\pm \frac{1}{x}$ only

A1F

6

(b) $1+2y = \left(4 - \frac{1}{x}\right)^2$

need $k \sqrt{1+2y} =$ 'x expression with + c'
and attempt to square both sides

m1

$$2y = 15 + \frac{1}{x^2} - \frac{8}{x}$$

terms on RHS in any order
AG – convinced CSO

A1

2

[8]

Q19.

(a) $\int \frac{dx}{\sqrt{x-6}} = \int -2dt$

Attempt to separate and integrate

M1

$$2\sqrt{x-6} = -2t + c$$

c on either side

A1A1

$$t = 0 \quad x = 70 \quad \Rightarrow \quad c = 16$$

Follow on c from sensible attempt at integrals

($\sqrt{\quad}$ not ln)

m1A1F

$$t = 8 - \sqrt{x-6}$$

CAO (orAEF)

A1

6

- (b) (i) The liquid level stops falling/flowing/at minimum depth

B1

1

(ii) $x = 22 \quad t = 8 - \sqrt{22 - 6}$

*Use $x = 22$ in their equation provided there is a c
Or start again using limits*

(M1) $2\sqrt{64} - 2\sqrt{16} = \pm 2t$, (A1) $t = 4$

M1

$t = 4$

CAO

A1

2

[9]

Q20.

$$\int \frac{dy}{y^2} = \int 6x \, dx$$

*Attempt to separate
Either dx or dy in right place*

M1

$$-\frac{1}{y} = 3x^2 (+C)$$

$$-\frac{1}{y}; 3x^2$$

A1A1

$x = 2 \quad y = 1 \quad C = -13$

Use (2, 1) to find a constant.

M1

CAO

A1

$$y = \frac{1}{13 - 3x^2}$$

CAO OE

A1

6

[6]

Q21.

(a) (i) $t = 0$ $x = 85$

B1

1

(ii) $t = 30$ $x = 15 + 70e^{-\frac{30}{40}}$

M1

$= 48.06 \approx 48^\circ\text{C}$

A1

2

(iii) $x = 60$

$$\frac{60 - 15}{70} = e^{-\frac{t}{40}}$$

$$\ln\left(\frac{45}{70}\right) = -\frac{t}{40}$$

Their 60 - 15

M1

EXAM PAPERS PRACTICE

$t = 17.67 \approx 18$ minutes

A1

3

(b) (i) $\int \frac{dx}{x - 15} = -\int \frac{dt}{40}$

Attempt to separate and integrate

M1

Correct expression and used

A1

$\ln(x - 15) = -\frac{t}{40} (+c)$

A1A1

$$(0, 85) \quad c = \ln 70$$

m1

$$\frac{t}{40} = \ln 70 - \ln(x - 15) \quad t = 40 \ln \left(\frac{70}{x - 15} \right)$$

Use (0.85) to find c , which must now appear in expression
Manipulate to $t = \dots$

A1

6

$$(ii) \quad e^{\frac{t}{40}} = \frac{70}{x - 15}$$

Manipulate expression including a $\ln$ towards $x = \dots$

M1

$$x - 15 = 70e^{-\frac{t}{40}}$$

AG; convincingly obtained

M1A1

2

[14]

Q22.

(a) Using $F = ma$:

$$-4v^{\frac{1}{3}} = 12 \frac{dv}{dt}$$

B1

$$\therefore \frac{dv}{dt} = -\frac{1}{3} v^{\frac{1}{3}}$$

$$-3 \int \frac{dv}{v^{\frac{1}{3}}} = \int dt$$

condone -, 3 incorrect side

M1

$$-3 \times \frac{v^{\frac{2}{3}}}{\frac{2}{3}} = t + c$$

$$-\frac{9}{2}v^{\frac{2}{3}} = t + c$$

condone lack of + c

A1

When $t = 0$, $v = 8 \Rightarrow c = -18$

M1A1

$$-\frac{9}{2}v^{\frac{2}{3}} = t + 18$$

$$v^{\frac{2}{3}} = 4 - \frac{2}{9}t$$

$$v = \left(4 - \frac{2}{9}t\right)^{\frac{3}{2}}$$

A1

6

(b) Particle is at rest when $4 - \frac{2}{9}t = 0$

The value of t is 18

B1

1

[7]

Q23.

(a) Using $F = ma$

$$1600 \frac{dv}{dt} = 4000 - 40v$$

$$\frac{dv}{dt} = \frac{4000 - 40v}{1600}$$

M1

$$\frac{dv}{dt} = \frac{100 - v}{40}$$

A1

2

(b) $40 \frac{dv}{100-v} = dt$

B1

$$40 \int \frac{dv}{100-v} = \int dt$$

M1

$$-40 \ln(100-v) = t + c$$

Condone lack of '+ c'

A1

When $t = 0, v = 0 \Rightarrow c = -40 \ln 100$

$$-40 \ln(100-v) = t - 40 \ln 100$$

$$t = 40 \ln \frac{100}{100-v}$$

$$e^{\frac{t}{40}} = \frac{100}{100-v}$$

M1A1

$$v = 100 - 100 e^{-\frac{t}{40}} \text{ or } 100(1 - e^{-\frac{t}{40}})$$

EXAM PAPERS PRACTICE

A1

6

©2025 Exam Papers Practice. All rights reserved.

[8]

Q24.

(a) using $F = ma$

$$0.4 \frac{dv}{dt} = 2 - 4v$$

M1

$$\frac{dv}{dt} = -10(v - 0.5)$$

Needs line above

A1

2

(b) hence $\int \frac{1}{v-0.5} dv = - \int 10 dt$
 $\ln(v-0.5) = -10t + c$

M1 for any side integrated correctly

M1A1

m1 for + c (and M1 gained)

m1

$v - 0.5 = Ce^{-10t}$
 $t = 0, v = 1$

$\therefore C = 0.5$

A1

$\therefore v = 0.5 + 0.5e^{-10t}$

condone $v = 0.5 + e^{-10t-0.693}$

A1

5

(c) when $v = 0.55$, $0.55 = 0.5 + 0.5e^{-10t}$

substitute 0.55 into C's (b), after finding c, possible numerical error

M1

$10 = e^{10t}$
 $t = \ln 10 \div 10$

EXAM PAPERS PRACTICE

A1

$= 0.230$

©2025 Exam Papers Practice. All rights reserved.

A1

3

[10]

Q25.

(a) Using $F = ma$:

$m \frac{dv}{dt} = 49 - 9.8v$ or $5g - 9.8v$

Need to see $m \frac{dv}{dt}$ or $5 \frac{dv}{dt}$ or $a = \frac{49-9.81}{5}$

M1

$$\therefore \frac{dv}{dt} = -1.96(v - 5)$$

Must see m terms (not a = ...)

A1

2

(b) $\int \frac{dv}{v-5} = -1.96 \int dt$

And one side integrated

M1

$$\ln(v - 5) = -1.96t + c$$

Need + c, A1 each side

A1A1

When $t = 0$, $v = 7 \Rightarrow c = \ln 2$

OE

A1

$$\ln \frac{v-5}{2} = -1.96t$$

$$\frac{v-5}{2} = e^{-1.96t}$$

$$v = 5 + 2e^{-1.96t}$$

CAO

A1

5

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[7]

Q26.

(a) (i) $F = 65g - 260v$

Accept $260v - 65g$

$$= 65(9.8 - 4v)$$

AG must see $65v$ or 260

B1

1

(ii) Using $F = ma$

$$65 \frac{dv}{dt} = 65(9.8 - 4v)$$

Need to see terms in m (condone – sign)

M1

$$\frac{dv}{dt} = -4(v - 2.45)$$

AG

A1

2

(b) $\frac{1}{v - 2.45} \frac{dv}{dt} = -4$

B1

$$\int \frac{1}{v - 2.45} dv = -\int 4 dt$$

$$\ln(v - 2.45) = -4t + c$$

M1: log side correct

M1

$$-4t + c$$

A1

$$v - 2.45 = Ce^{-4t}$$

$$t = 0, v = 19.6$$

$$\therefore C = 17.15 \text{ or } e^{2.84}$$

$$\text{Or } c = \ln 17.15 \text{ or } 2.84$$

A1

$$\therefore v = 2.45 + 17.15e^{-4t} \quad 2.45 + 17.2e^{-4t}$$

A1

5

[8]

Q27.

(a) Using $F = ma$

$$-2mv\frac{5}{4} = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -2v^{\frac{5}{4}} \quad \text{AG}$$

B1: Must see $-2mv^{\frac{5}{4}} = m \frac{dv}{dt}$ or $-2mv^{\frac{5}{4}} = ma$ and correct final answer.

B1

1

(b) $\int \frac{dv}{v^{\frac{5}{4}}} = -2 \int dt$

M1: Two integrals with one in the form $\int f(v)dv$ where $f(v) = v^{\pm \frac{5}{4}}$ or $v^{\pm \frac{4}{5}}$. The other integral must not contain v terms.

M1

$$-\frac{4}{v^{\frac{1}{4}}} = -2t + c$$

A1: Correct expression.
Condone lack of $+ c$ for this A1, but no subsequent marks if no c .

A1

When $t = 0$, $v = 16 \Rightarrow c = -2$

dM1: Using $t = 0$ and $v = 16$ to find c .

A1: Obtaining $c = -2$.

dM1

A1

$$-\frac{4}{v^{\frac{1}{4}}} = -2t - 2$$

$$v^{\frac{1}{4}} = \frac{2}{1+t}$$

$$v = \left(\frac{2}{1+t} \right)^4 \quad \text{AG}$$

A1: Correct final answer. Must see $v^{\frac{1}{4}} = \frac{2}{1+t}$

or $v^{-\frac{1}{4}} = \frac{1+t}{2}$ or $\frac{1}{v^{\frac{1}{4}}} = \frac{1+t}{2}$
Or

if they obtain $v = \left(\frac{2}{t+c} \right)^4$

$v = 16, t = 0 \Rightarrow 16^{\frac{1}{4}} = \frac{2}{c}$, condone $c = 1$ (no other root considered)

A1

5

[6]

Q28.

(a) Using $F = ma$,

$$-0.2mv^{\frac{1}{2}} = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$

AG Must see equ'n containing m

B1

1

(b) $\int \frac{dv}{v^{\frac{1}{2}}} = -\int 0.2 dt$

$$2v^{\frac{1}{2}} = -0.2t + c$$

$m1$ for $+c$

M1

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

A1m1

When $t = 0, v = 16 \therefore C = 8$

A1

$$2v^{\frac{1}{2}} = -0.2t + 8$$

$$v = (4 - 0.1t)^2$$

AG

A1

5

(c) When $v = 1, 1 = (4 - 0.1t)^2$
 $4 - 0.1t = \pm 1$

M1

$$t = 30 \text{ or } 50$$

$$\left[\text{if use } 2v^{\frac{1}{2}} = 8 - 0.2t \text{ no need to see } 50 \right]$$

A1

$$t = 30$$

$t \neq 50$ as ball stops when $t = 40$

A1

3

(d) Integrating $v = (4 - 0.1t)^2$

$$v = 16 - 0.8t + 0.01t^2$$

$$x = 16t - 0.4t^2 + \frac{0.01}{3}t^3 + d$$

M1 for first 3 terms or $-\frac{10}{3}(4 - 0.1t)^3$

M1

When $t = 0, x = 0 \Rightarrow d = 0$

A1

$$x = 16t - 0.4t^2 + \frac{0.01}{3}t^3$$

When speed is $1 \text{ m s}^{-1}, t = 30$

$$x = 480 - 360 + 90$$

dep on M1 above

m1

$$= 210$$

[No 'd', 3 marks only]

A1

4

[13]

Q29.

$$\frac{dv}{dt} = -\frac{\lambda}{v^4}$$

M1

$$\int v^{\frac{1}{4}} dv = -\int \lambda dt$$

Condone one of $v^{-\frac{1}{4}}, + \int \lambda dt, \frac{1}{\lambda}$

m1

$$\frac{4}{5} v^{\frac{5}{4}} = -\lambda t + c$$

m1 for + c

A1A1m1

$$t = 0, v = u \therefore c = \frac{4}{5} u^{\frac{5}{4}}$$

A1

$$\therefore v^{\frac{5}{4}} = u^{\frac{5}{4}} - \frac{5}{4} \lambda t$$

$$v = \left(u^{\frac{5}{4}} - \frac{5}{4} \lambda t \right)^{\frac{4}{5}}$$

A1

[7]

Q30.

- (a) Using $F = ma$:
 $-0.08v^2 = 0.05a$

B1

$$\therefore \frac{dv}{dt} = -1.6v^2$$

AG; condone sign error in first B1

B1

2

- (b) $\int \frac{dv}{v^2} = -1.6 \int dt$

M1

$$-\frac{1}{v} = -1.6t(+c)$$

Condone $-\frac{1}{v} = -1.6t + c \Rightarrow \frac{1}{v} = 1.6t + c$

A1

When $t = 0, v = 3 \Rightarrow c = -\frac{1}{3}$

M1

$$\frac{1}{v} = \frac{1}{3} + 1.6t \quad *$$

A1

$$\begin{aligned} \frac{1}{v} &= \frac{1}{3} + \frac{8}{5}t \\ \frac{1}{v} &= \frac{5 + 24t}{15} \\ v &= \frac{15}{5 + 24t} \end{aligned}$$

AG; all working lines correct from *

A1

5

[7]

Q31.

(a) Using $F = ma$:

$$-\lambda m v^{\frac{3}{2}} = m \frac{dv}{dt}$$

M1

$$\therefore \frac{dv}{dt} = -\lambda v^{\frac{3}{2}}$$

AG

A1

2

(b) $\int \frac{dv}{v^{\frac{3}{2}}} = -\lambda \int dt$

M1

$$-\frac{2}{v^{\frac{1}{2}}} = -\lambda t + c$$

Condone no '+c'

A1

When $t = 0, v = 9 \Rightarrow c = -\frac{2}{3}$

Dep. on correct integration

(accept sign or ' $\frac{1}{2}$ ' error)

M1A1

$$\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3}$$

A1

$$\frac{\sqrt{v}}{2} = \frac{1}{\lambda t + \frac{2}{3}}$$

$$v = \left(\frac{6}{2 + 3\lambda t} \right)^2$$

Needs correct algebra

m1

$$v = \frac{36}{(2 + 3\lambda t)^2}$$

AG

A1

7

(c) When $v = 4$,

$$\text{or } \frac{36}{(2 + 3\lambda t)^2} = 4 \quad M1$$

$$\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3} \Rightarrow 1 = \lambda t + \frac{2}{3}$$

$$(2 + 3\lambda t)^2 = 9 \quad A1$$

M1A1

$$t = \frac{1}{3\lambda}$$

$$t = \frac{1}{3\lambda} \text{ A1 needs statement why}$$

$$2 + 3\lambda t \neq -3$$

A1

3

[12]

Q32.

(a) Using $F = ma$

$$-0.05mv = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.05v$$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05 dt$

B1

$$\ln v = -0.05t + c$$

$$v = Ce^{-0.05t}$$

Need first 2 terms

M1

$$\text{When } t = 0, v = 20$$

$$\therefore C = 20$$

$$v = 20e^{-0.05t}$$

only for fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$$e^{0.05t} = 2$$

A1

$$\therefore t = \frac{1}{0.05} \ln 2$$

$$= 13.9 \text{ or } 20 \ln 2$$

A1

3

- (d) Work done by force is change in KE of car

M1

$$= \frac{1}{2} m (20)^2 - \frac{1}{2} m (10)^2$$

A1

$$= 150 \text{ m}$$

A1

3

[11]

Q33.

- (a) Using $F = ma$

$$-0.05mv = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.05v$$

Need to see m terms

B1

1

- (b) $\int \frac{dv}{v} = -\int 0.05 \, dt$

B1

$$\ln v = -0.05t + c$$

$$v = Ce^{-0.05t}$$

Need first 2 terms

M1

$$\text{When } t = 0, v = 20$$

$$\therefore C = 20$$

$$v = 20e^{-0.05t}$$

fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$$e^{0.05t} = 2$$

A1

$$\therefore t = \frac{1}{0.05} \ln 2$$

$$= 13.9$$

Accept 20 ln 2

A1

3

[8]

Q34.

(a) Using $F = ma$:

$$-\lambda mv = ma = m \frac{dv}{dt}$$

Condone no '-'

M1

$$\therefore \frac{dv}{dt} = -\lambda v$$

AG

Note: no use of $m \Rightarrow$ no marks in (a)

A1

2

(b) $\int \frac{dv}{v} = -\lambda \int dt$

M1

$$\ln v = -\lambda t + c$$

$$v = C e^{-\lambda t}$$

Needs '+ c'

A1

When $t = 0$, $v = U \Rightarrow C = U$

Needs correct working

M1

$$v = U e^{-\lambda t}$$

AG

A1

4

[6]

Q35.

$$1600 \frac{dv}{dt} = -40v$$

Applying Newton's second law with $40v$ and $\frac{dv}{dt}$

M1

Correct equation

A1

$$\int \frac{1}{v} dv = \int -\frac{1}{40} dt$$

Separating variables

dM1

$$\ln v = -\frac{t}{40} + c$$

integrating to get $\ln v$ term.

dM1

$$v = Ae^{-\frac{t}{40}}$$

Correct integral with or without c

A1

$$t = 0, v = 20 \Rightarrow c = 20$$

Finding constant

dM1

$$v = 20e^{-\frac{t}{40}}$$

Correct final result

A1



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.